

Known

$$F.S = 3$$

$$\delta = .3 \text{ mm}$$

$$P = 300 \text{ N}$$

$$d = 18 \text{ mm}$$

PETG



Unknown

Thickness

Max Stress

$$\delta = \frac{ML^2}{2EI}$$

$$M = F \cdot r \cdot F.S$$

$$I = \frac{bh^3}{12}$$

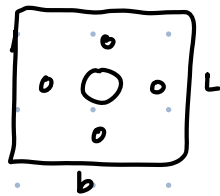
$$G \geq h$$

$$b \geq 28 \text{ mm}$$

$$E = 26 \text{ GPa}$$

$$L \geq 28 \text{ mm}$$

Assume a $30 \times 30 \text{ mm}$ Plate



$$b_{\text{thin}} = 30 - 6 - 2(3) = 18 \text{ mm}$$

$$\delta = \frac{ML^2}{2E \frac{bh^3}{12}} \rightarrow \delta = \frac{6ML^2}{Eb^3h}$$

$$h = \left[\frac{6ML^2}{Eb\delta} \right]^{1/3}$$

$$\sigma_y = \frac{M_y}{I} = \frac{M \left(\frac{h}{2} \right)}{\frac{bh^3}{12}} = \frac{6M}{bh^2} \xrightarrow{\text{Solve for } h} h = \sqrt{\frac{6M}{b\sigma_y}}$$

$$M = F \cdot r \cdot F.S$$

$$M = (300)(.012)(3)$$

$$M = 10.8 \text{ N}\cdot\text{m}$$

$$h_B = \left[\frac{(6)(10.8)(.03)^2}{(2 \times 10^9)(.018)(.0003)} \right]^{1/3} = .017544 \text{ m}$$

or

$$17.544 \text{ mm}$$

$$h_\sigma = \left[\frac{(6)(10.8)}{(1.018)(5 \times 10^7)} \right]^{1/2} = .008485 \text{ m}$$

or

$$8.485 \text{ mm}$$

Known

$$\delta = .3 \text{ mm}$$

$$F.S. = 3$$

$$4 \text{ holes} = 3.4 \text{ mm}$$

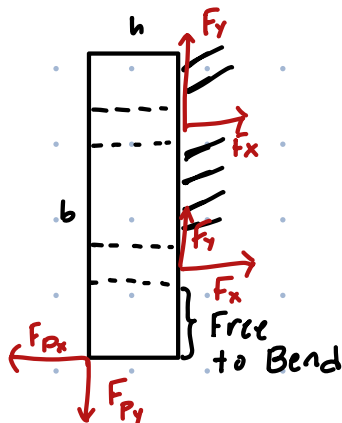
$$A = 30 \times 47.544 \text{ mm}^2$$

$$P = 300$$

PET6

Unknown

Max thickness
Moment



$$\delta = \frac{ML^2}{2EI} \rightarrow \delta = \frac{ML^2}{2E \frac{bh^3}{12}} \rightarrow \delta = \frac{6ML^2}{Eb^3}$$

$$h_b = \left[\frac{6ML^2}{Eb\delta} \right]^{1/3}$$

$$\sigma_y = \frac{M_y}{I} = \frac{M \left(\frac{h}{2} \right)}{\frac{bh^3}{12}} = \frac{6M}{bh^2} \stackrel{\text{Solve for } h}{=} h_\sigma = \sqrt{\frac{6M}{b\sigma_y}}$$

$$M = (300)(.012 + .017554)(3)$$

$$h_b = \left[\frac{(6)(26.5986)(.017554)^2}{(2 \times 10^9)(.03)(.0003)} \right]^{1/3} = .013980 \text{ m}$$

or

13.980 mm

$$h_\sigma = \left[\frac{(6)(26.5986)}{(.03)(5 \times 10^7)} \right]^{1/2} = .010315 \text{ m}$$

or

10.315 mm

